Auditing Financial ML Backtests with Evaluation Contracts and Executable Checks

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Abstract

A backtest turns portfolio decisions into investment results through rules for execution, valuation, cash income, and trading costs. In the reference implementation, we separate portfolio construction from return accounting to examine how these rules affect model evaluation. Each accounting comparison uses the same portfolio decisions and prices. New holdings first earn returns over the interval after execution. We revisit a fund rotation strategy and compare gradient boosting with simple trading rules across sector and country funds. Applying a decision to a return interval that ended before execution can raise or lower performance and reverse strategy rankings. Cash income and trading costs also change the comparisons, while the model seed affects the position of gradient boosting among the rules. For the rotation strategy, our return decomposition shows that allocation and trading costs outweigh positive contributions from passive exposure and exposure timing. Both studies use market histories available during development. We resample consecutive observations to obtain pointwise intervals conditional on the fitted models, portfolio decisions, and seeds in each study.

1 Introduction

Consider a model that chooses a portfolio after observing today's close. Its holdings earn returns after execution. Crediting the new allocation with today's return changes the historical experiment, as does leaving idle capital at zero return or charging just one side of a trade. These choices enter the performance statistic used to judge the model.

We evaluate allocations to exchange-traded funds, or ETFs. A target weight states the desired fraction of capital in an asset. The backtest determines when that target becomes a holding, how prices change its weight, and what cash and trading contribute to wealth. Statistical inference must then account for dependence and the way the strategy entered the study. Repeated search through a limited history creates a risk of false discoveries in financial machine learning [White, 2000, Harvey et al., 2016, Bailey and López de Prado, 2014, Bailey et al., 2017, Arnott et al., 2019].

We define an evaluation contract as the accounting rules and checks that connect portfolio decisions to returns. Each model or trading rule supplies an allocation record with the decision time, asset symbol, and target weight. Running the same sequence of decisions under different timing, cash, and cost conventions isolates their consequences for performance. We decompose daily return above cash into contributions from allocation, exposure timing, passive exposure, and trading costs.

The first experiment revisits a weekly ETF rotation strategy that predates this audit. The second follows a protocol written before data retrieval and compares gradient boosting with time series momentum, cross-sectional momentum, and short-term reversal on two ETF sets. Five random seeds show how the regression model's portfolio performance varies.

Our contribution is an audit of how financial model performance is measured. We trace returns from the information available at a decision through execution, cash flows, and costs, then examine

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how the evaluation conventions affect model comparisons. Checks on information availability and portfolio accounting support that analysis. Separate order tests examine submission state under simulated failures.

2 Related work

Evaluation validity: Adaptive reuse of a test set changes the inferential problem [Dwork et al., 2015]. A benchmark also measures a particular task, whose relationship to a broader capability needs justification [Raji et al., 2021]. In scientific applications of machine learning, leakage can give evaluation access to information that prediction would lack [Kapoor and Narayanan, 2023]. Our object of measurement is historical portfolio performance under explicit allocation and accounting rules.

Financial backtest validity: Sharpe inference depends on assumptions about distributions and serial dependence [Lo, 2002]. Searching many strategies on the same history can overstate the performance of the one ultimately chosen [White, 2000, Bailey and López de Prado, 2014, Bailey et al., 2017, Harvey et al., 2016]. Arnott et al. [2019] organize these risks into a research protocol, and Zhang et al. [2026] study contamination at decision time across models and equity datasets. We examine a timing error in which a portfolio decision receives the return from an interval that ends before execution is possible.

Attribution and comparison: Performance analysis has long separated decisions into return components [Fama, 1972, Brinson and Fachler, 1985]. Our exact arithmetic decomposition separates daily return above cash into passive exposure, ex post exposure timing, active allocation, and modeled trading costs. The basket specified below provides the reference for allocation and exposure. The target rules draw on momentum, reversal, residual momentum, and gradient boosting [Jegadeesh and Titman, 1993, Moskowitz et al., 2012, Lehmann, 1990, Blitz et al., 2011, Friedman, 2001]. The expansion's momentum and reversal rules are long-only ETF adaptations; their windows and weights are specified below. The retrospective rule scores ETF residuals from a regression on one factor. Blitz et al. construct stock portfolios from residuals of the Fama and French model with three factors: market, size, and value. Their study motivates the use of residual signals. Our ETF universe, regression on one factor, and allocation rule differ from theirs.

Checks and reproducibility: CheckList tests natural language processing behaviors beyond aggregate accuracy [Ribeiro et al., 2020]. The checks in this paper address financial evaluation rules defined here. Qlib and FinRL provide modular workflows for financial AI [Yang et al., 2020, Liu et al., 2020], while FinRL-X joins data, strategy construction, backtesting, and broker execution through a weight interface [Yang et al., 2027]. Reproducing submitted financial analyses from the same raw inputs is itself difficult [Pérignon et al., 2024]. Yin et al. [2026] compare five engines across 15 strategies. Our engine comparison reuses the same allocation records while changing daily weight resets, weight drift, trade accounting, and sizing after costs together. The NeurIPS 2019 reproducibility program combined code, community reproduction, and a checklist [Pineau et al., 2021].

3 Evaluation contract

An evaluation rule is useful when its implementation can be checked. We apply design by contract to portfolio invariants [Meyer, 1992]: a run stops when an input or accounting quantity violates a required condition. The contract defines the experiment; the engine computes holdings, cash, costs, and returns under those rules. Figure 1 follows this calculation, and Table 1 lists the checks and their evidence.

3.1 Portfolio accounting

A target decided at a close earns returns only on later intervals. Let $w_d \in \mathbb{R}^N$ be the target for risky assets decided after observing the close at decision time d, and let $e(d)$ be the first available execution bar strictly after d. We express holdings in accounting units valued at adjusted closing prices. At

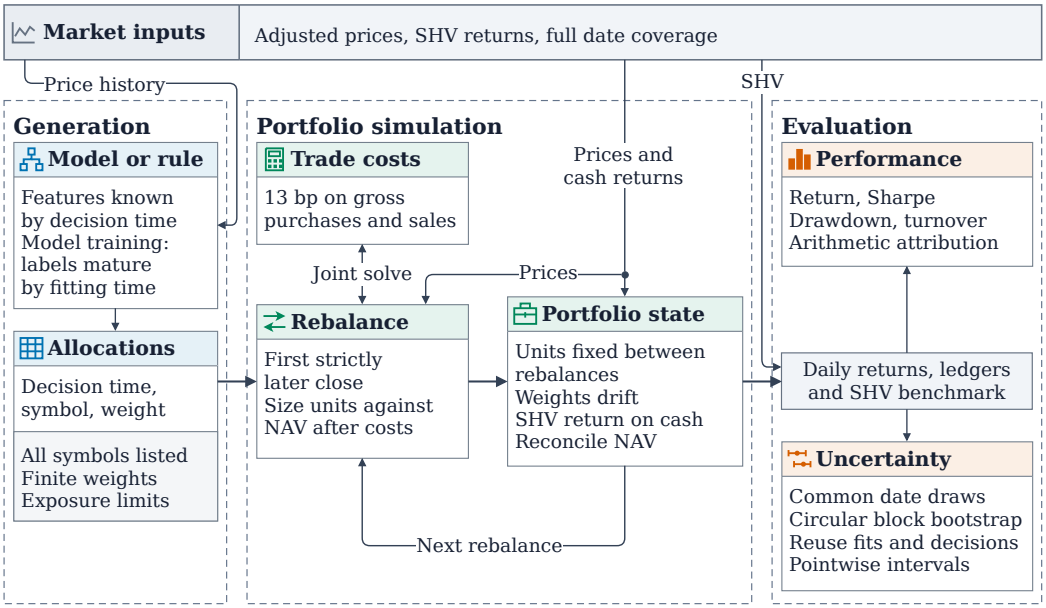


Figure 1: Baseline evaluation with the event engine. Generation limits features by decision time and training labels by fitting time. The engine solves for trade quantities and costs together. Fees reduce NAV at execution; new units earn returns over the next interval. Timing, cash, and cost comparisons rerun the same targets under each convention, then align both return paths with SHV. Performance statistics and circular block resampling use these daily records. Resampling keeps the fitted models, portfolio decisions, and seeds unchanged and recomputes metrics to form pointwise intervals.

$e(d)$, the primary event engine solves for quantities $q_{i,e(d)}$ against net asset value after costs, denoted NAV, so that the resulting risky allocation matches w_d. These quantities remain fixed between rebalances, while portfolio weights drift as prices and NAV change. If $P_{i,t}$ is the adjusted closing price and V_t is portfolio NAV, the risky weight at the prior close is $h_{i,t-1} = q_{i,t-1}P_{i,t-1}/V_{t-1}$. The resulting holdings first earn returns over the interval following the execution close. For the long-only case, $h_{i,t} \geq 0$ and $\sum_i h_{i,t} \leq 1$; residual cash is

$$h_t^c = 1 - \sum_{i=1}^{N} h_{i,t}. \tag{1}$$

For simple asset returns $r_{i,t}$ over $(t-1, t]$ and return on the cash account r_t^c, portfolio return before costs is

$$r_t^{\text{gross}} = \sum_{i=1}^{N} h_{i,t-1} r_{i,t} + h_{t-1}^c r_t^c. \tag{2}$$

The engine rejects gaps in a supplied series of cash returns. For the change in executed risky notional $\Delta a_{i,t}$, measured against NAV before trading, define $B_t = \sum_i \max(\Delta a_{i,t}, 0)$ and $S_t = \sum_i \max(-\Delta a_{i,t}, 0)$. One-way turnover is $\max(B_t, S_t)$. The charge applies to purchases plus sales, $B_t + S_t$, at a rate of 13 basis points: $\kappa_t = 0.0013(B_t + S_t)$ of NAV before trading. If V_t^- is NAV before trading and V_{t-1} is NAV at the prior close, the reduction in period return is $d_t = \kappa_t V_t^- / V_{t-1}$ and $r_t^{\text{net}} = r_t^{\text{gross}} - d_t$. The engine solves for accounting quantities using NAV after costs. We use this flat charge to study sensitivity to trading. Estimating execution cost also depends on spread, market impact, order size, liquidity, volatility, and venue information [Almgren and Chriss, 2001]. The same proportional schedule applies to every trade in these experiments.

Table 1: Evaluation rules and the checks or records used to assess them.

Area	Rule	Check or evidence
Allocation record	Decision timestamp, symbol, and target weight; each decision is a complete long-only book within the gross limit.	Schema and runtime checks reject records with invalid fields, incomplete books, and invalid weights.
Information and timing	Features must be available by their decision times, and training labels when fitting begins. Execution uses the first later bar.	The generator checks feature and label availability. The evaluator checks execution alignment.
Cash and cost	Residual cash earns the return calculated from adjusted closing prices of the Treasury ETF SHV; 13 basis points apply to gross notional bought and sold.	Checks reject missing cash returns and absent cost models; ledgers reconcile.
Comparison	Basket return at actual exposure; a separate portfolio follows target exposure and pays its own costs.	Daily return reconciliation, reproducible comparisons, and cost records.
Engines	Accounting quantities stay fixed between trades in the event engine; the vectorized engine resets weights daily.	Common target weights reveal differences between the accounting conventions.
Order state	Record intent before submission. Prevent a retry after an uncertain outcome and a new identifier while an intent for the same symbol remains unresolved.	Simulated failures exercise stale writes, retries, and replacement identifiers in the order ledger.

We report nominal compound annual growth rate, or CAGR, maximum drawdown, annualized one-way turnover, gross traded notional, and the sample Sharpe ratio relative to benchmark returns r_t^f:

$$\mathrm{SR}_f = \sqrt{252}\,\frac{\mathrm{mean}(r_t - r_t^f)}{\mathrm{sd}(r_t - r_t^f)}. \tag{3}$$

The baseline sets $r_t^c = r_t^f = r_t^{\mathrm{SHV}}$. The comparison with zero cash income sets $r_t^c = 0$, while the convention comparisons retain $r_t^f = r_t^{\mathrm{SHV}}$. The cash comparison changes the portfolio's income while using the same reporting benchmark. Equation 3 gives the conventional descriptive annualization. Its use for inference depends on assumptions about distributions and serial dependence [Lo, 2002]. We use the circular block bootstrap [Politis and Romano, 1992]: each draw samples sequences of consecutive dates with replacement, and wraps from the last observation to the first when a sequence crosses the sample boundary. These sequences, or blocks, retain dependence among their observations.

We credit cash with returns calculated from SHV closing prices after the provider's adjustments. SHV is a Treasury ETF, so this convention includes fund expenses, duration exposure, and changes in its market price. Risky holdings use adjusted prices as well. The resulting units serve total return accounting; implementing these positions as share purchases would also need market and execution records.

3.2 Return decomposition

We decompose net return in excess of cash into allocation, timing, passive exposure, and cost. Let $x_t = \sum_i h_{i,t-1}$ be risky exposure during return period t, let b_t be the return of a buy-and-hold basket with equal initial weights, and let $\bar{x}$ be the mean of x_t over the full sample. Basket returns at actual exposure and at mean exposure are

$$m_t = x_t b_t + (1 - x_t) r_t^c, \tag{4}$$

$$k_t = \bar{x} b_t + (1 - \bar{x}) r_t^c. \tag{5}$$

Then net excess over cash has the exact arithmetic decomposition

$$r_t^{\text{net}} - r_t^c = \underbrace{(r_t^{\text{gross}} - m_t)}_{\text{active risky allocation}} + \underbrace{(m_t - k_t)}_{\text{dynamic exposure timing}} + \underbrace{(k_t - r_t^c)}_{\text{passive risky exposure}} + \underbrace{(r_t^{\text{net}} - r_t^{\text{gross}})}_{\text{implementation cost}}. \tag{6}$$

The allocation term combines group selection, weights within the risky portfolio, and drift between rebalances. Its reference is the basket return at the strategy's actual exposure. Brinson selection instead weights return differences within groups by benchmark group weights. These definitions answer different attribution questions.

The timing term compares exposure on each day with its mean over the evaluation window. It is positive when exposure is above that mean and the basket outperforms cash, or when exposure is below the mean and the basket underperforms cash. The mean uses the full sample, so the timing term describes the return history ex post. Interpreting the terms as causal effects or as returns on separately tradable portfolios involves assumptions beyond this accounting decomposition.

Averaging the daily components and multiplying by 252 gives their annualized arithmetic contributions. These contributions add to annualized mean return above cash. CAGR instead compounds the daily net returns, so it has a different aggregation rule.

We resample the five series in Equation 6 together, using the same sequences of consecutive dates. Their component means add to mean net return above cash in every resample. Each component's interval uses its own percentiles of the resampling distribution. Adding these component bounds generally gives different endpoints from the interval for mean net return above cash. The fraction of positive resamples describes the distribution of the estimated mean.

3.3 Accounting comparisons

Three comparisons reuse the same portfolio decisions, input prices, and primary engine. Each changes one convention: cash earns zero, the proportional trading charge is removed, or a new allocation receives the return interval ending at its decision close. The last convention assigns a past return to a decision made after observing that close. It is deliberately infeasible and tests the effect of incorrect return timing.

Cash income and trading charges change NAV, which can affect subsequent exposure and trade size even when the target weights stay the same. Our reruns include those accounting consequences. Following Zhang et al. [2026], each of these comparisons changes one convention. The basket and engine comparisons described below change several aspects of the calculation together.

3.4 Validation checks

The checks cover allocation sums, position limits, exposure after overlays, and information availability. Each evaluation starts from a complete allocation record. In both studies, the strategy's decisions also determine the comparison portfolio's exposure, letting us examine how different allocations and accounting rules turn those decisions into returns.

Synthetic examples with answers calculated by hand check the accounting independently of market data. The experiment records identify the inputs, settings, and software environment needed to repeat each calculation. Separate order tests simulate stale state, an uncertain submission outcome, and a replacement order for the same symbol. The submission record must retain the earlier intent and prevent a second submission until the outcome is reconciled. Appendix J distinguishes this test evidence from the checks needed for live execution.

4 Evaluation design

4.1 Retrospective audit

The first study evaluates a weekly rotation strategy on six ETFs from January 2, 2018, through December 31, 2025. It combines group ranking, residual momentum, exposure states from a Gaussian

Table 2: Portfolio rules in the expansion study. Gradient boosting fits regression trees, abbreviated GBRT in the plots. Ties use ascending symbols. Reversal and gradient boosting assign one-third to each qualifying asset, leaving the unused allocation in cash when fewer than three qualify.

Family	Portfolio rule
Time series momentum	Positive simple returns over 252 sessions; equal weight on qualifying assets; hold cash if none qualify.
Cross-sectional momentum	Three highest simple returns from $t - 252$ to $t - 21$; one-third each.
Short-term reversal	Up to three lowest negative simple returns over 5 sessions; one-third each, without renormalizing.
Gradient boosting	Up to three highest strictly positive predictions of log return over 21 sessions; one-third each, without renormalizing; five seeds.

mixture, and volatility scaling. The strategy predates the audit, and the number of earlier trials is unknown. We retain that history as the object of the accounting investigation.

The original strategy contains a defect that matters for interpreting its returns. QQQ is both a group member and the ranking benchmark. Its regression on itself should give zero residuals, but positive rounding error can survive the score filter and receive substantial weight after normalization. Two regime features are exact rescalings of the same volatility series, early regime windows contain little fitting history, and one seed determines the mixture fit. We preserve these choices, including the QQQ defect, to audit the strategy's original portfolio decisions. Appendix B gives the rule.

The event engine credits cash with SHV returns, charges 13 basis points on purchases and sales, and executes each weekly allocation at the first later close. The decomposition uses the return of a basket at the strategy's actual exposure before costs. A separate comparison portfolio starts with equal basket weights, lets their composition drift, follows the strategy's target exposure, and pays its own trading charges. Both portfolios use the same execution timing and cash convention. This comparison measures the contribution of allocation choices after costs along the strategy's exposure schedule.

4.2 Expansion study

We wrote the second study's protocol before requesting its data. Because the evaluation period had already occurred, this chronology limits discretion over the analysis of a historical sample. It does not create an independent test period. Section 6 discusses historical reuse and the absence of external registration.

Two ETF sets use adjusted closing prices from January 2, 2014, through December 31, 2025. The sector set contains XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, and XLY. The country set contains EWA, EWC, EWG, EWH, EWJ, EWL, EWP, EWQ, EWS, and EWU. Each also includes SHV. Evaluation covers 2,011 sessions from January 2, 2018, through December 31, 2025. Each dataset keeps only sessions with observations for every required symbol. Preprocessing excludes forward filling and symbol substitution. The sets consist of surviving funds with shared market exposures, so their results describe related parts of the same history.

Table 2 gives the target rules. Each monthly decision uses the first complete session and executes on the first strictly later close. The fraction allocated to risky assets is at most one, with short positions excluded; idle capital earns SHV returns. The event engine holds quantities valued at adjusted closing prices fixed between rebalances and charges 13 basis points for every dollar bought or sold, using the $B_t + S_t$ cost base defined above. The evaluator checks the target weights before calculating returns. The comparator preserves target gross risky exposure, divides it equally across the risky ETFs, excluding SHV, at each decision, then executes at the first later close, credits residual cash with SHV returns, and pays its own trading charges.

At each monthly decision, we fit one regressor per seed to observations pooled across the risky assets in that ETF set. A training row is eligible once its return outcome can be observed at that close. Features use prices available by their own decision dates. Appendix H gives the feature windows, return label, training window, model settings, and seeds.

Each seed produces its own allocation record and portfolio returns. We calculate metrics separately for those portfolios, then average the five values. The gradient boosting Sharpe entry is the average of the five individual ratios. A portfolio that combines their holdings would be a different investment strategy.

4.3 Comparison design

Five comparisons use the same dates and the subtraction directions below. Three change one accounting convention while reusing the allocation records: retroactive close assignment minus execution at the first later close, zero return on residual cash minus SHV cash, and zero trading charge minus the charge of 13 basis points. The fourth subtracts the performance of the comparison portfolio after its own costs from that of the net strategy, with both executing at the first later close and using the same target exposure. The fifth subtracts performance under event engine accounting from performance under vectorized accounting. The comparator changes cross-sectional allocation, turnover, and cost. The engine comparison combines daily weight resets, weight drift, trade accounting, and sizing after costs. Each comparison uses a separate rerun. Runs combining the changes would measure their joint effect. If the target on the first evaluation date is nonzero, retroactive assignment charges its initial entry before the evaluation window; execution at the first later close charges it inside the window. The timing contrast also includes that difference in the timing of entry costs.

Primary outcomes are differences in annualized arithmetic return and sample Sharpe relative to SHV. We construct unstudentized pointwise 95% percentile intervals from 5,000 circular block resamples of aligned daily observations. The primary block contains 21 consecutive sessions; sensitivity checks use 5 and 63 sessions [Politis and Romano, 1992]. Within each family and ETF set, both sides of a comparison and SHV use the same sampled dates. For gradient boosting, those dates are shared by all five seed pairs. Model fits, ETF sets, families, allocation records, design choices, and seeds are held unchanged during resampling. Each interval describes one comparison under those conditions. Section 6 addresses multiplicity and the uncertainty that this resampling leaves outside its scope.

5 Results

5.1 Performance across ETF sets

Sector ETFs outperform country ETFs on arithmetic return, compound growth, and Sharpe relative to SHV for all four strategy families over this period. Country time series momentum illustrates why the cash benchmark matters: its nominal return is positive, but its return relative to SHV gives a slightly negative Sharpe ratio. Capital growth alone misses that opportunity cost.

The risk comparison between gradient boosting and cross-sectional momentum reverses across ETF sets. For country ETFs, the regression portfolios have lower mean compound growth, and every seed has lower volatility and a shallower drawdown than the momentum rule. For sector ETFs, every seed has higher volatility and a deeper drawdown. The model's risk profile depends on the investment universe even with the same fitting procedure. Appendix Table 6 reports the eight cases, and Figure 4 compares return and Sharpe estimates. Model entries average metrics from separate seed portfolios.

5.2 Effects of accounting conventions

Retroactive close assignment has its largest effect on arithmetic return for short-term reversal. The rule selects assets with negative recent returns using a window that includes the decision close. Assigning those assets to the interval ending at that close credits the portfolio with a return that helped select them. The observed direction is consistent with that mechanism: annualized arithmetic return falls by about seven percentage points in each ETF set. Both pointwise intervals from blocks of 21 sessions lie below zero. Sector gradient boosting is the third case with an interval below zero among the eight timing comparisons in Appendix Figure 5.

Time series momentum moves in the opposite direction: retroactive assignment raises its return in both sets, although both pointwise intervals include zero. Together, the opposite signs show how a timing error can produce an apparently conservative result. Feasibility depends on the order of

Table 3: Sample Sharpe relative to SHV and rank within each ETF set. Rank one is highest. Bold marks the highest sample Sharpe within each set and convention. Each alternative changes timing or trading cost separately from the baseline. Retroactive assignment is infeasible and can move the initial entry charge outside the evaluation window. Zero trading cost retains baseline timing and cash income. Gradient boosting entries average five individual seed Sharpes.

	Baseline		Retroactive assignment *infeasible*		Zero trading cost	
Strategy	Sharpe ratio	Rank	Sharpe ratio	Rank	Sharpe ratio	Rank
United States sectors						
Time series momentum	0.494	3	0.526	2	0.525	4
Cross-sectional momentum	**0.596**	1	**0.572**	1	**0.630**	1
Short-term reversal	0.439	4	0.055	4	0.531	3
Gradient boosting	0.529	2	0.409	3	0.606	2
Country equities						
Time series momentum	−0.018	4	0.044	3	0.022	4
Cross-sectional momentum	**0.329**	1	**0.313**	1	0.364	2
Short-term reversal	0.197	3	−0.186	4	0.297	3
Gradient boosting	0.314	2	0.244	2	**0.413**	1

observation, decision, execution, and return, regardless of which direction the performance statistic moves. Appendix A.1 illustrates both signs with two assets.

Cash effects depend on how much capital remains uninvested and when. Cross-sectional momentum stays almost fully invested and changes little when cash earns zero. Reversal holds more cash, especially in countries, and loses more income under that convention. Five of the eight pointwise cash intervals lie below zero. SHV remains the reporting benchmark in both runs, so the comparison captures cash income and its subsequent effects on NAV and holdings.

Removing trading charges raises annualized arithmetic return by 0.57 to 1.81 percentage points, with all eight pointwise intervals above zero. Differences in the amount traded explain much of the variation in cost effects across strategies. Country reversal trades substantially more gross notional per year than sector time series momentum, and multiplying traded notional by the charge approximates the corresponding return reduction. The full comparison measures sensitivity to this proportional cost assumption by rerunning the portfolios, since charges also change NAV and later trade sizes.

5.3 Strategy rankings

Accounting conventions also change the sample ranking of the strategies in Table 3. Retroactive assignment swaps gradient boosting with time series momentum in sectors, and reversal with time series momentum in countries. Removing trading costs swaps reversal with time series momentum in sectors and puts gradient boosting above cross-sectional momentum in countries. Each alternative produces one pairwise swap in each ETF set.

Some ranking changes reflect small Sharpe differences. With zero cost, sector reversal and time series momentum differ by about 0.006, which disappears when both ratios are rounded to two decimal places. We retain three decimals in the table for that reason. These sample rankings describe the observed portfolios. Appendix Figure 6 instead gives pointwise intervals for the change in each strategy's Sharpe when its accounting convention changes.

5.4 Sources of return

The comparison portfolios follow each strategy's target exposure, execution timing, and cash convention, but replace its risky allocation with a basket that pays its own trading costs. The retrospective basket starts with equal weights and then drifts; the expansion basket returns to equal weights at every decision. These comparisons ask whether the allocation rule improves performance along the same intended exposure schedule after trading costs. Seven of the eight expansion intervals for

annualized arithmetic return include zero. Country time series momentum is the exception, with a negative difference from its basket and a pointwise interval below zero. Positive nominal returns can coexist with little evidence of an allocation advantage.

Cash treatment is especially consequential for the retrospective strategy. It spends 46.89% of the sample entirely in cash. Its compound growth falls below SHV despite that cash income and falls further when cash earns zero. The arithmetic decomposition shows a corresponding shortfall in mean return above cash: active allocation and trading costs outweigh positive contributions from passive exposure and exposure timing.

The pointwise interval for the allocation contribution lies below zero. Appendix Table 5 reports all component estimates and intervals; their means add to net return above cash before rounding. This accounting decomposition identifies where the observed shortfall arises under the chosen basket reference.

The comparison with a basket after its own costs answers a different question. The decomposition holds the strategy's actual exposure when attributing allocation return; the comparison portfolio follows target exposure and incurs its own trades and charges. Differences in exposure and cost treatment explain why its return comparison differs from the allocation component. Appendix B.1 reports both calculations.

5.5 Sensitivity checks

Appendix Figure 7 compares the regression portfolios for the five seeds with the simple trading rules. The mean Sharpe summarizes the five portfolios under one training design. Evaluating a portfolio that combines their holdings would require calculating its own return series and Sharpe ratio.

The mean can conceal differences in model ranking. In sectors, four of the five regression runs exceed time series momentum on Sharpe relative to SHV, but only one exceeds cross-sectional momentum. In countries, every run exceeds time series momentum and two exceed cross-sectional momentum. Those two also have higher CAGR than cross-sectional momentum, even though every regression run has lower annualized arithmetic mean return. Sharpe uses both the mean and standard deviation of returns above cash; CAGR compounds daily returns. Model comparisons depend on the metric and seed within this shared historical sample. Appendix I examines convention effects, annual excess returns, and trading for each seed.

The engine comparison changes how portfolio decisions become daily holdings. Both engines receive the same target weights and prices. The vectorized calculation resets weights to the last executed vector each day; the event engine keeps quantities unchanged between trades, so weights move with prices. The engines also differ in trade accounting and sizing after costs. Appendix Figure 8 reports the largest absolute daily return difference for each family and ETF set.

The largest absolute daily return difference occurs for sector gradient boosting, while sector cross-sectional momentum has the largest difference in final wealth, 3.89% of initial capital. We calculate final wealth by compounding each engine's return series separately. The largest daily difference and the final wealth difference measure different aspects of engine behavior, since daily differences can accumulate or offset one another.

The retrospective strategy's returns agree much more closely across the engines. Engine sensitivity depends on both the portfolio decisions and the accounting conventions.

Changing the resampling block length leaves the interval conclusions for timing, cost, and the basket comparisons unchanged across 5, 21, and 63 sessions, as summarized in Appendix Table 7. The small sector gradient boosting cash effect includes zero with the longest blocks. For engine differences, the treatment of temporal dependence changes the interval conclusions: two intervals for annualized arithmetic return lie above zero with blocks of 5 sessions, compared with four for blocks of 21 and 63 sessions.

5.6 Performance across subperiods

The retrospective strategy has higher CAGR but lower Sharpe relative to SHV in 2022 through 2025 than in 2018 through 2021. The comparison basket exceeds the strategy on both measures in each

half, so its advantage over the full sample also appears in both subperiods. These comparisons divide the original return history without changing its portfolio decisions.

6 Limitations

Both studies use historical periods available during development. Earlier trials of the rotation strategy are incompletely recorded, and the expansion protocol remains under author control without external registration. The comparisons establish accounting effects for these strategies on these histories. Assessing performance on an independent sample means setting aside a period throughout development or collecting future observations.

The ETF sets share dates and market exposures, and the five seeds repeat one model specification. Pointwise intervals describe resampling of daily returns conditional on the models and portfolio decisions already used. Nominal coverage relies on stationarity, sufficiently weak dependence, and suitable moment conditions. These intervals exclude uncertainty from model selection and from sampling a wider population of seeds. Simultaneous coverage requires a multiplicity procedure. The incomplete search history also prevents calculation of a Deflated Sharpe Ratio or a White Reality Check [White, 2000, Bailey and López de Prado, 2014]. Appendix A explains these limits and the dependence assumptions.

Historical accounting and live execution provide different kinds of evidence. Capacity and execution quality depend on liquidity, order size, market impact, and venue conditions beyond the flat charge used here. Simulated order tests check particular failure scenarios; broker behavior during live operation needs separate observation. Treating these conditional comparisons as evidence for live deployment could expose investors to losses from underestimated risk or trading costs. This audit helps a reader inspect the assumptions behind a performance claim before using it to inform an investment decision.

7 Conclusion

A model's portfolio decisions acquire a performance history through execution and accounting rules. Our comparisons show that an infeasible timing convention can raise or lower measured performance, while cash income and trading intensity also affect it. The retrospective decomposition traces the shortfall relative to cash to allocation and costs despite positive exposure and timing contributions. Interpreting model performance requires this connection between the information available at the decision, the holdings that follow execution, and the cash flows that produce the return.

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A Scope of the evidence

Adjusting inference for strategy search requires a record of the trials. The Deflated Sharpe Ratio uses the effective number of independent trials and the variance across trial Sharpe estimates, together with the selected return series' Sharpe ratio, length, skewness, and kurtosis. White's Reality Check requires a candidate universe, benchmark, performance criterion, and observations suitable for joint resampling or equivalent retained summaries [White, 2000, Bailey and López de Prado, 2014]. The available trial history is incomplete, so we report sample metrics and conditional intervals.

The allocation component can also be written as $A_t = \sum_i h_{i,t-1}(r_{i,t} - b_t)$, a contribution per unit of portfolio capital. This expression remains defined when the portfolio holds only cash. For the timing component, let $T_t = m_t - k_t$, $z_t = b_t - r_t^c$, and let $\bar{z}$ denote the mean over the n evaluation sessions. Then

$$252\bar{T} = \frac{252}{n}\sum_{t=1}^{n}(x_t - \bar{x})(z_t - \bar{z}). \tag{7}$$

The mean timing contribution is positive when exposure and basket excess return move together over the sample. Its annualized mean equals 252 times their empirical covariance, calculated with denominator n. The basket at mean exposure uses an ex post reference; a causal timing interpretation or a separately tradable timing strategy requires additional assumptions. The bootstrap resamples the precomputed components while keeping the original $\bar{x}$ fixed, so every draw uses the same reference.

The decomposition describes arithmetic excess return; geometric linking needs a separate construction. The fraction of bootstrap draws with a positive mean is a frequency within the resampling distribution. Posterior probabilities need a Bayesian probability model, and inference across multiple comparisons needs an appropriate testing procedure.

The datasets contain funds available at the time of collection. Point-in-time constituent histories would change the investment opportunity set. Country ETFs are unhedged securities listed in the U.S., so currency and overlapping market exposures enter their returns. SHV contributes duration, expenses, and variation in market prices to the cash proxy. The quantities valued at adjusted closing prices are units for total return accounting. Executable share quantities, spreads, market impact, volume constraints, and fill uncertainty would require additional market and broker records.

The regression model uses one specification and five fixed seeds. It pools asset observations within each ETF set without using the symbol as a feature. The bootstrap conditions on the resulting fitted models and target weights. Nominal coverage relies on stationarity, sufficiently weak dependence, suitable moment conditions, and an appropriate block construction. We compare three block lengths to assess interval sensitivity. This comparison does not test the assumptions on the return process. The uncertainty estimates exclude model selection, hyperparameter search, population variation across seeds, and future performance.

The audit tests particular invariants and failure cases. Estimating a rate of defect detection would need a defined population of faulty implementations, and measuring the incremental value of a check would need comparisons with procedures that omit it. The convention and overlay comparisons measure changes in portfolio accounting and exposure.

The simulated order tests exercise stale state, uncertain retries, and replacement identifiers. The order ledger stops a new order while an earlier intent for the same symbol remains unresolved, so even a valid later order may wait for reconciliation. Establishing pending exposure across the account, delivery semantics across concurrent clients, authenticated transport, and venue behavior requires further operational evidence. These are prerequisites for assessing a funded deployment.

A.1 A worked example of retroactive assignment

Consider two assets observed at four successive closes, with prices given below. Starting wealth is 1, held entirely in cash, and both cash return and trading charge are zero. After observing t_1, a winner rule selects A, while a loser rule selects B; either rule targets full investment in the selected asset. These rules illustrate one interval. The empirical time series momentum and reversal rules use 252 and 5 sessions, respectively.

Close	Price of A	Price of B	Feasible chronology
t_0	100	100	Entirely in cash
t_1	110	90	Observe prices and select A or B
t_2	110	90	Execute the selected target
t_3	110	90	First return earned by the new holding

Feasible execution occurs at t_2. The new holding first earns the return ending at t_3, which is $110/110 - 1 = 0$ for A and $90/90 - 1 = 0$ for B. Both feasible portfolios therefore finish with wealth 1.00.

Retroactive assignment instead credits the already ended interval from t_0 to t_1: $110/100 - 1 = +0.10$ for A or $90/100 - 1 = -0.10$ for B. Subsequent prices are unchanged, so the fictitious terminal wealth is 1.10 or 0.90.

The same chronology error creates a gain for the winner rule and a loss for the loser rule. This example isolates return timing. The empirical comparison uses a trading charge of 13 basis points and includes any shift of the initial entry charge across the evaluation boundary.

B Retrospective case specification

The universe is QQQ, VGT, GLD, TLT, SPY, and VIG, grouped as growth equities, gold and nominal duration, and broad and dividend equities. On the first session of each week, the strategy keeps the top two groups by an information ratio over 126 sessions relative to QQQ. Within selected groups, it estimates a regression on one factor over 126 sessions, then sums residual returns over the next 126 sessions, leaving the most recent 21 sessions out of score formation. Each residual equals the asset return minus the fitted intercept and benchmark contribution, and enters the score in return units with coefficient one. Positive scores receive proportional long-only weights capped at 60% per asset.

A Gaussian mixture with three components and full covariance uses seed 42, covariance regularization 10^{-5}, and at most 100 iterations. It maps QQQ return and two exact rescalings of realized volatility over 20 sessions to zero, half, or full exposure. A separate volatility scaler uses 21 sessions, targets 20 volatility points, and clips its multiplier to $[0.3, 1.0]$. The strategy requires 274 sessions of history before evaluation; 503 are available. If a benchmark, group signal, or regression estimate is unavailable, it falls back to plain momentum. It holds cash if all mature residual scores in the selected groups are nonpositive. The comparison evaluates all named variants: removing the regime rule, removing the volatility scaler, and removing both.

B.1 Retrospective accounting results

Table 4: Retrospective performance under baseline accounting from 2018 through 2025. Every Sharpe in the table uses SHV as its reporting benchmark. The path with zero return on residual cash retains SHV as the reporting benchmark. Its Sharpe against zero is 0.11. The comparison portfolio executes at the first later close and matches the strategy's target exposure.

Series	Compound annual growth rate	Sharpe ratio	Maximum drawdown
Strategy, after costs, with SHV cash	2.05%	-0.04	-9.25%
Strategy, before costs, with SHV cash	3.86%	0.25	-8.51%
SHV cash proxy	2.50%	Undefined	-0.45%
Basket at actual exposure, before costs	7.11%	0.85	-6.21%
Comparison portfolio, after costs	6.10%	0.67	-6.22%
Strategy, after costs, with zero cash return	0.47%	-0.31	-10.59%

After each portfolio pays its own trading costs, the annualized arithmetic mean of daily strategy returns minus comparison returns is -3.86%. Resampling those differences in circular blocks of 21 sessions gives a pointwise 95% percentile interval from -6.40% to -1.35%, conditional on the observed portfolio decisions.

Table 5: Exact attribution of annualized arithmetic net excess return over SHV. Pointwise intervals use common resamples of blocks of 21 sessions. The daily decomposition holds in each draw, and component means sum to net excess before rounding. Values are rounded to two decimals.

Component	Mean	95% interval	Draws with a positive mean
Active risky allocation	-3.05%	[-5.52%, -0.57%]	35/5000
Dynamic exposure timing	0.54%	[-2.35%, 3.45%]	3182/5000
Passive risky exposure	4.00%	[0.85%, 7.04%]	4963/5000
Modeled implementation cost	-1.76%	[-2.12%, -1.41%]	0/5000
Net excess return over cash	-0.26%	[-4.29%, 3.82%]	2340/5000

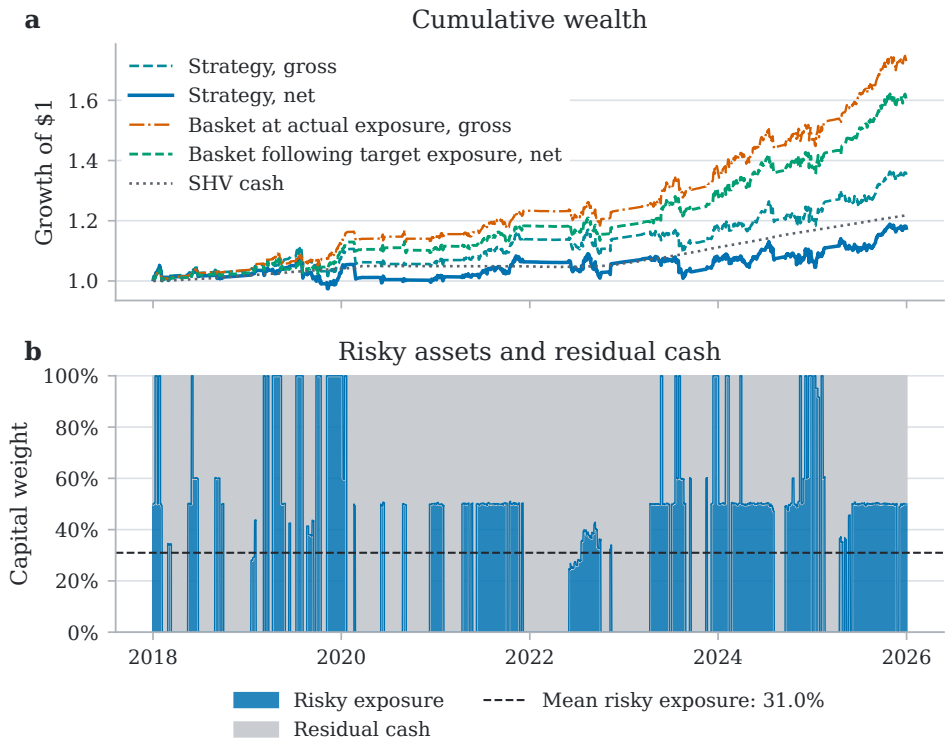


Figure 2: Growth of one dollar and allocation to risky assets and residual cash in the retrospective study. Exposure is indexed by the ending date of each return interval. Cash earns SHV returns. Residual cash accounts for most capital on average.

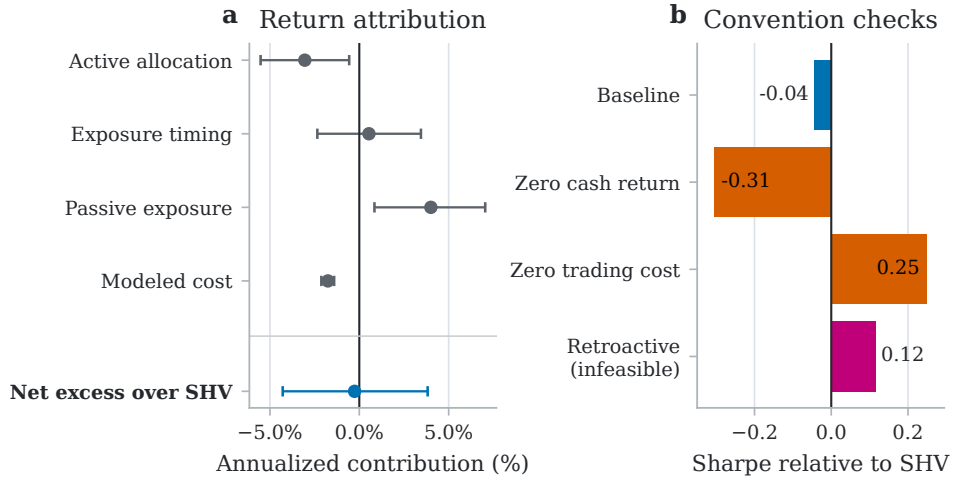


Figure 3: Retrospective attribution and convention checks. Subplot a reports annualized arithmetic contributions to excess return over SHV, with pointwise 95% intervals from common draws using circular blocks of 21 sessions. The separate bottom row sums the four component means and has its own interval from the common draws. Subplot b gives Sharpe levels relative to SHV under each convention. Hatching marks retroactive assignment, which credits a target with an interval that has already ended.

C Code and data availability

The public materials include the code, aggregate results, software environment specification, and synthetic accounting examples. The historical price matrices and empirical target weights are not distributed. Reproducing the reported returns requires access to the original inputs and the stated

software environment. Prices were retrieved from Yahoo Finance through yfinance; later downloads may reflect revisions to historical prices. Permission from the data provider to publish derived aggregates or redistribute source data remains unverified. The synthetic examples allow the accounting checks to run without market data.

D Expansion results

Table 6: Expansion performance after the proportional trading charge, with Sharpe ratios relative to SHV. Bold marks the highest arithmetic return, Sharpe ratio, and compound growth and the shallowest drawdown within each ETF set, using unrounded values. These are sample rankings; uncertainty in differences requires a separate comparison. For gradient boosting, each metric is calculated separately for all five seeds and then averaged. This order also applies to compound annual growth and maximum drawdown.

Fund set	Strategy	Annualized arithmetic return	Sharpe ratio	Compound annual growth rate	Maximum drawdown
United States sectors	Time series momentum	11.44%	0.49	10.29%	-34.16%
United States sectors	Cross-sectional momentum	**14.19%**	**0.60**	**13.04%**	**-32.06%**
United States sectors	Short-term reversal	11.02%	0.44	9.54%	-45.08%
United States sectors	Gradient boosting	13.66%	0.53	12.10%	-39.96%
Country equities	Time series momentum	2.18%	-0.02	0.88%	-40.45%
Country equities	Cross-sectional momentum	**8.84%**	**0.33**	**7.21%**	-40.73%
Country equities	Short-term reversal	6.03%	0.20	4.48%	-43.91%
Country equities	Gradient boosting	7.87%	0.31	6.60%	**-33.56%**

Table 7: Sensitivity to resampling block length. Each cell counts pointwise intervals below zero, including zero, and above zero, in that order, across eight dependent strategy and ETF set cases. Return differences are annualized arithmetic means; Sharpe differences use SHV as the benchmark. We vary the number of consecutive sessions per block while keeping the fitted models and portfolio decisions fixed. Simultaneous inference across cases requires a multiplicity adjustment.

Metric	5 sessions	21 sessions	63 sessions
Retroactive assignment minus execution at the first later close			
Arithmetic return	3/5/0	3/5/0	3/5/0
Sharpe ratio	3/5/0	3/5/0	3/5/0
Zero cash return minus SHV cash return			
Arithmetic return	5/3/0	5/3/0	4/4/0
Sharpe ratio	5/3/0	5/3/0	4/4/0
Zero trading cost minus a charge of 13 basis points			
Arithmetic return	0/0/8	0/0/8	0/0/8
Sharpe ratio	0/0/8	0/0/8	0/0/8
Net strategy minus comparison portfolio			
Arithmetic return	1/7/0	1/7/0	1/7/0
Sharpe ratio	1/7/0	1/7/0	1/7/0
Vectorized engine minus event-driven engine			
Arithmetic return	0/6/2	0/4/4	0/4/4
Sharpe ratio	0/7/1	0/4/4	0/4/4

Return and Sharpe can indicate different directions of change in the same comparison. With the primary blocks of 21 sessions in Figures 5 and 6, all 40 pairs of intervals occupy the same position relative to zero, and 39 pairs of point estimates share a sign. The exception is the sector regression model against its comparison portfolios after trading costs: its mean annualized arithmetic return is higher by 1.04 percentage points, but its mean Sharpe is lower by 0.023. Both intervals include zero.

For two return paths on the same dates, let $\mu_j = 252\ \mathrm{mean}(r_j - r^{\mathrm{SHV}})$ and $\sigma_j = \sqrt{252}\ \mathrm{sd}(r_j - r^{\mathrm{SHV}}) > 0$ for $j \in \{A, B\}$, using the sample standard deviation. Then

$$\mathrm{SR}_A - \mathrm{SR}_B = \frac{\mu_A - \mu_B}{\sigma_A} + \mu_B \left(\frac{1}{\sigma_A} - \frac{1}{\sigma_B} \right). \tag{8}$$

The common SHV series makes $\mu_A - \mu_B$ equal to the difference in annualized arithmetic returns. Equal standard deviations of returns above SHV preserve its sign; different standard deviations can change the ordering. For the regression model, the equality holds for each seed pair before averaging.

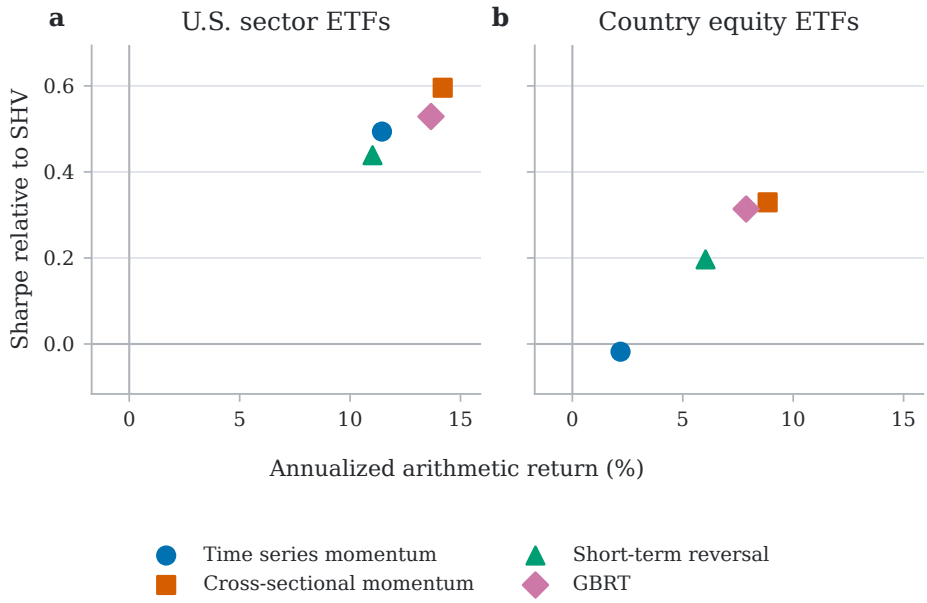


Figure 4: Nominal annualized arithmetic return and Sharpe relative to SHV after trading costs for the eight expansion cases. Each coordinate for the regression model is the arithmetic mean of that metric across five seeds, as defined in Table 6. Figure 7 shows the individual seed Sharpes alongside the three rules.

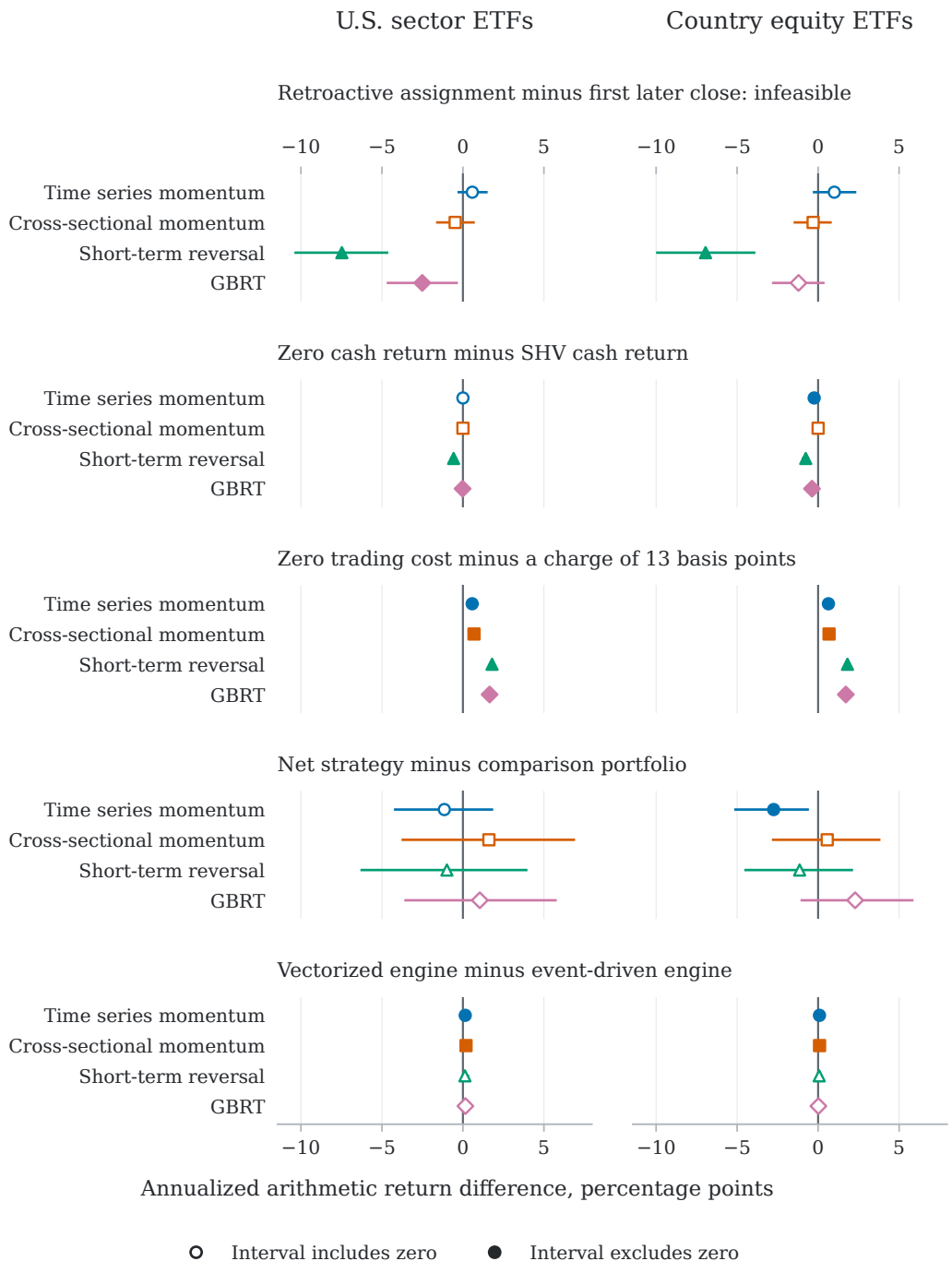


Figure 5: Changes in annualized arithmetic return, in percentage points. Group headings give the subtraction order. Lines show unstudentized pointwise 95% percentile intervals from common resamples of circular blocks of 21 sessions. Fitted models, target weights, and seeds stay fixed. Points for the regression model average differences calculated separately for each of the five seeds. Hollow markers identify intervals including zero, using unrounded bounds. Some intervals are narrower than their markers at this common scale. Retroactive assignment is infeasible and can move the initial entry charge outside the evaluation window. SHV remains the reporting benchmark when cash earns zero.

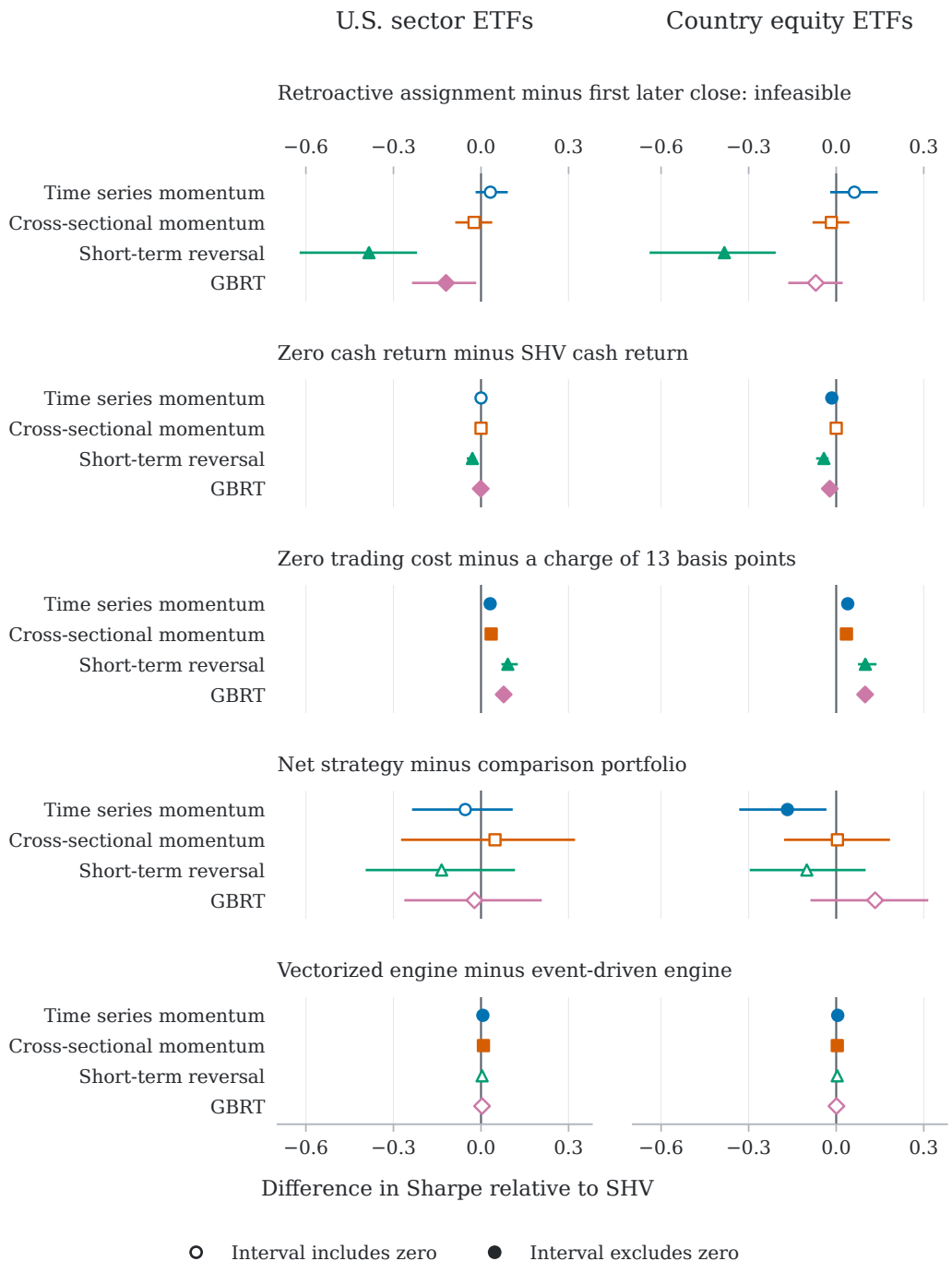


Figure 6: Changes in Sharpe relative to SHV under the five signed comparisons. Lines show unstudentized pointwise 95% percentile intervals from 5,000 common draws using circular blocks of 21 sessions. The draws vary aligned dates while retaining the fitted models, target weights, and five seeds. Points for the regression model average the five Sharpe differences calculated separately for each seed. Hollow markers identify intervals including zero, using unrounded bounds; some intervals are narrower than their markers. Retroactive assignment uses an infeasible execution chronology. Zero cash credit changes cash income while SHV remains the Sharpe benchmark.

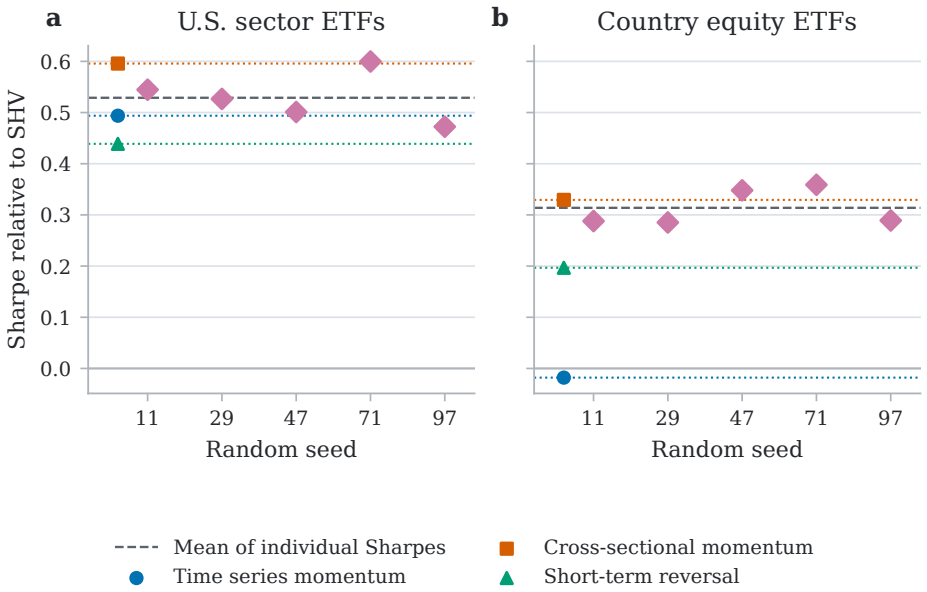


Figure 7: Baseline Sharpe relative to SHV for five model seeds in each ETF set. Diamonds show the regression model's Sharpe for each seed; dashed lines show the mean of the five Sharpes. Dotted lines and their leftmost markers show where each rule's Sharpe falls relative to the seed values. The spread describes these runs on shared history. Inference over a population of seeds requires a separate sampling design.

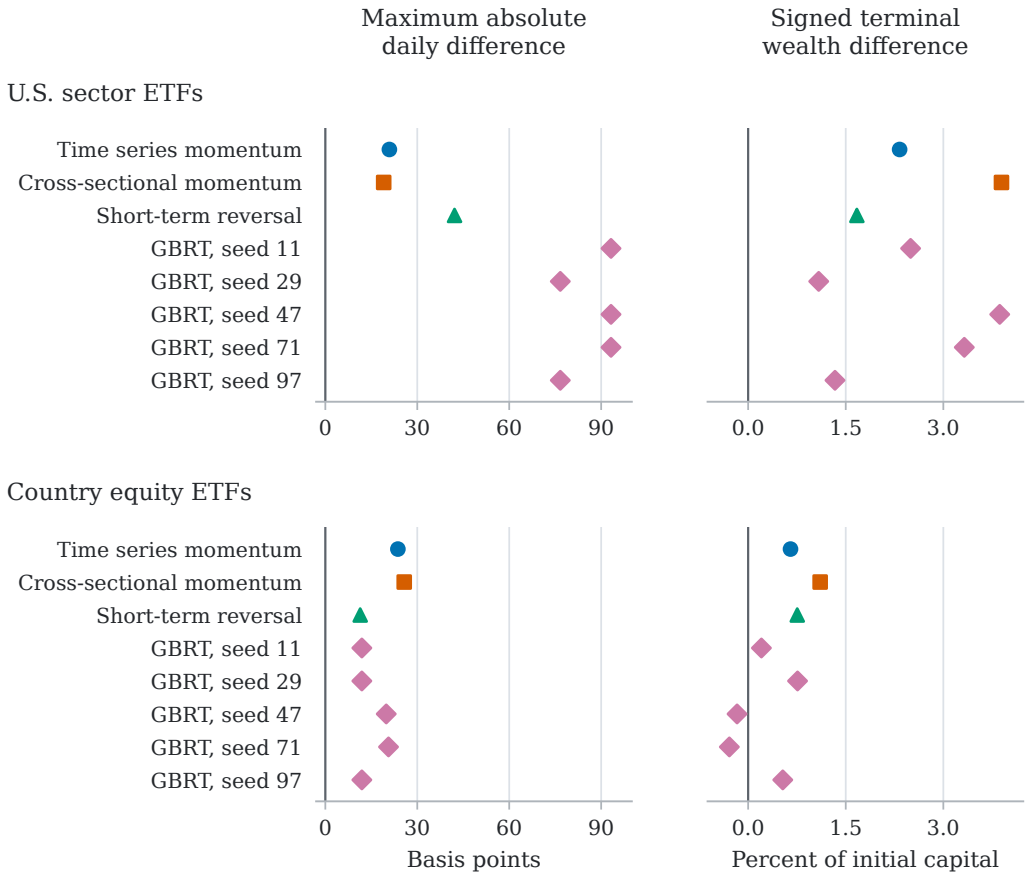


Figure 8: Engine differences for each rule and each seed of the regression model. Daily maxima measure the largest absolute return difference, in basis points. Terminal gaps subtract event engine wealth from vectorized wealth, as a percentage of initial capital. A large daily discrepancy can coexist with a small terminal gap as differences accumulate with opposing signs. The event engine holds accounting quantities fixed between trades; the vectorized convention resets weights daily. Trade accounting and sizing after costs also differ.

E Retrospective block length sensitivity

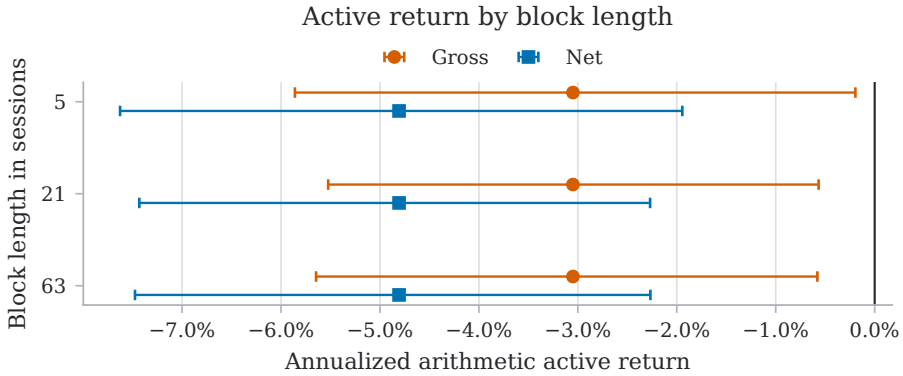
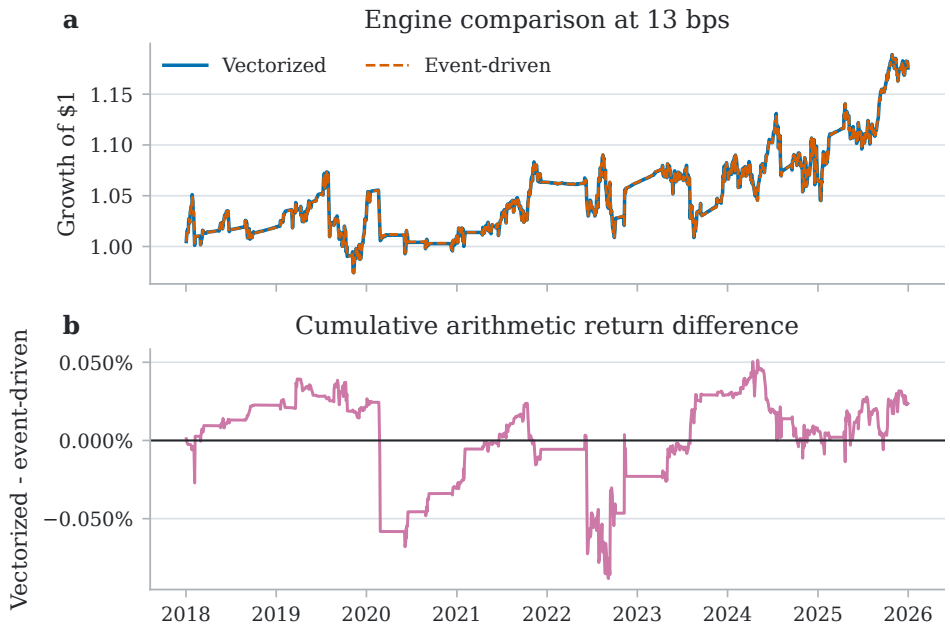


Figure 9: Annualized arithmetic active return relative to the basket at actual exposure before costs. Intervals are pointwise 95% bootstrap percentiles for circular blocks of 5, 21, and 63 sessions. Gross subtracts the basket return before costs from the strategy return before costs. Net deducts trading costs from the strategy return and uses the same basket return before costs.

F Retrospective engine comparison



Cumulative arithmetic difference: final +0.024%; maximum absolute 0.088%

Figure 10: Subplot a shows growth of one dollar under the two engines. Subplot b shows the arithmetic sum of daily return differences, subtracting event engine returns from vectorized engine returns. This sum differs from subtracting the compounded wealth paths in subplot a. The event engine holds accounting quantities valued at adjusted closing prices fixed between rebalances and allows portfolio weights to drift.

G Retrospective cost and overlay sensitivity

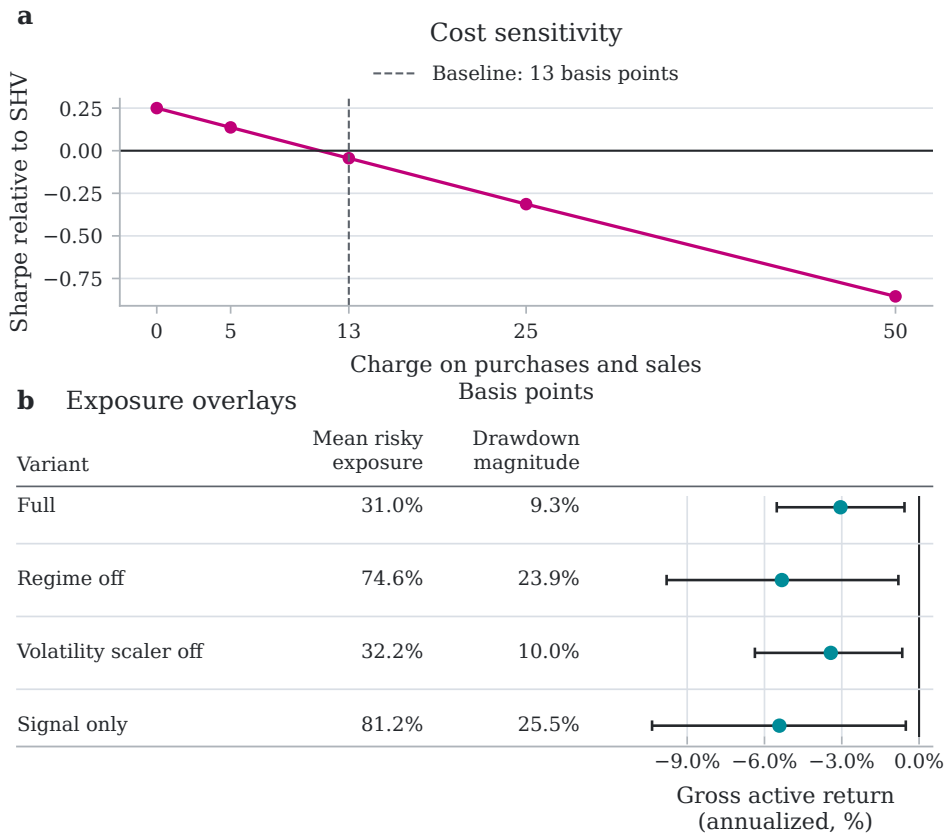


Figure 11: Retrospective sensitivity to trading costs and exposure overlays. Subplot a gives Sharpe relative to SHV under proportional charges on purchases and sales. Subplot b gives gross annualized arithmetic active return relative to the basket at each variant's actual exposure before costs, with pointwise 95% intervals from common draws using circular blocks of 21 sessions. The columns give mean risky exposure and maximum net drawdown magnitude. Each interval describes a variant's level relative to its own basket; an interval for an overlay effect would require a difference between variants.

The overlays change how much capital is exposed to risky assets. Removing the regime rule raises mean risky exposure and deepens the maximum drawdown; removing both overlays makes these changes larger. All four intervals for gross active return lie below zero relative to the basket at each variant's actual exposure. The comparisons retain the retrospective generator defect and measure the joint consequences of changes in exposure and the return path.

Positive compound growth can coexist with a shortfall against cash. In the cost grid, Sharpe relative to SHV falls from 0.14 at 5 basis points per dollar bought or sold to -0.04 at 13 basis points. CAGR remains positive at 0.40% under a charge of 25 basis points and falls to -2.93% at 50 basis points. Complete accounting reruns on the same target weights measure sensitivity to the five assumed charges, with SHV retained as the Sharpe benchmark.

H Expansion target specification

Time series momentum keeps every asset with a positive simple return over 252 sessions and assigns equal weight among those assets. If none qualify, the target is cash. Cross-sectional momentum ranks simple returns from session $t - 252$ to session $t - 21$, breaks ties by ascending symbol, and assigns one-third to each of the three assets with the highest returns. The rule fills all three slots even when a selected return is negative.

Short-term reversal ranks simple returns over 5 sessions and keeps up to three assets with the lowest strictly negative returns. Each receives one-third of capital. Any unused slot remains in cash, so a single qualifying asset receives the same weight as one member of a selection of three assets.

A single gradient boosting specification produces the regression model's target weights through the same allocation record interface as the three rules. Each record contains the decision time, symbol, and target weight.

For each decision, we calculate and store features from prices available at that close. The features comprise log returns over 5, 21, 63, 126, and 252 sessions, the sample standard deviation of daily log returns over 21 and 63 sessions, the price divided by its high over the trailing 252 sessions minus one, and ordinal cross-sectional ranks of log returns over 21 and 252 sessions. The ranks use the scale $[0, 1]$. The model uses these features without symbol indicators, standardization, or winsorization. The label is the log return over the following 21 sessions. It starts at the decision close, one close before portfolio entry, so the forecast window differs from the executable holding period.

At each monthly decision, we fit a separate model for each seed to asset observations pooled within the ETF set. Training includes only rows whose label endpoint falls on or before that close. It keeps up to the latest 60 decision months with mature labels and requires at least 24. Each fit uses 100 trees of depth 2, learning rate 0.03, minimum leaf size 10, subsample 0.8, and squared error loss, under a limit of one thread. The seeds are 11, 29, 47, 71, and 97.

After fitting, the generator keeps up to three assets with the highest strictly positive predictions, breaks ties by ascending symbol, and assigns one-third to each. The remainder stays in cash. A perturbation test changes prices after a cutoff and checks that this generator's target at the cutoff stays unchanged. Each additional generator needs its own checks on information availability.

I Historical variation within the expansion study

These descriptive analyses were added during revision after inspection of the aggregate results. They retain the original ETF sets, target weights, model settings, and accounting rules. Outcomes for each calendar year and the fitting paths for the five fixed seeds describe temporal variation and variation among model runs, respectively [Bouthillier et al., 2021].

I.1 Performance by calendar year

For each baseline run and year y, we calculate

$$A_y = \frac{252}{n_y} \sum_{t \in y} \left(r_t^{\text{net}} - r_t^{\text{SHV}} \right), \tag{9}$$

where n_y counts the observed sessions in that year. The slices retain the holdings, returns across year boundaries, and costs from the continuous evaluation. Weighting the annual values by their session counts recovers the full sample annualized arithmetic excess return, $\sum_y n_y A_y / \sum_y n_y$. Figure 12 shows every calendar year in the evaluation window. Entries for the regression model average the five individual seed values.

The ordering over the full sample combines different annual outcomes on the same history. All four country strategies have negative annualized excess returns over SHV in 2018 and 2022, and positive values in 2019, 2021, 2023, and 2025. The sector strategies diverge in 2022: cross-sectional momentum and reversal earn positive arithmetic returns above SHV, while the mean for the regression model is negative. Gradient boosting leads the sector set in 2021 and 2024; cross-sectional momentum leads in 2020, 2022, and 2025.

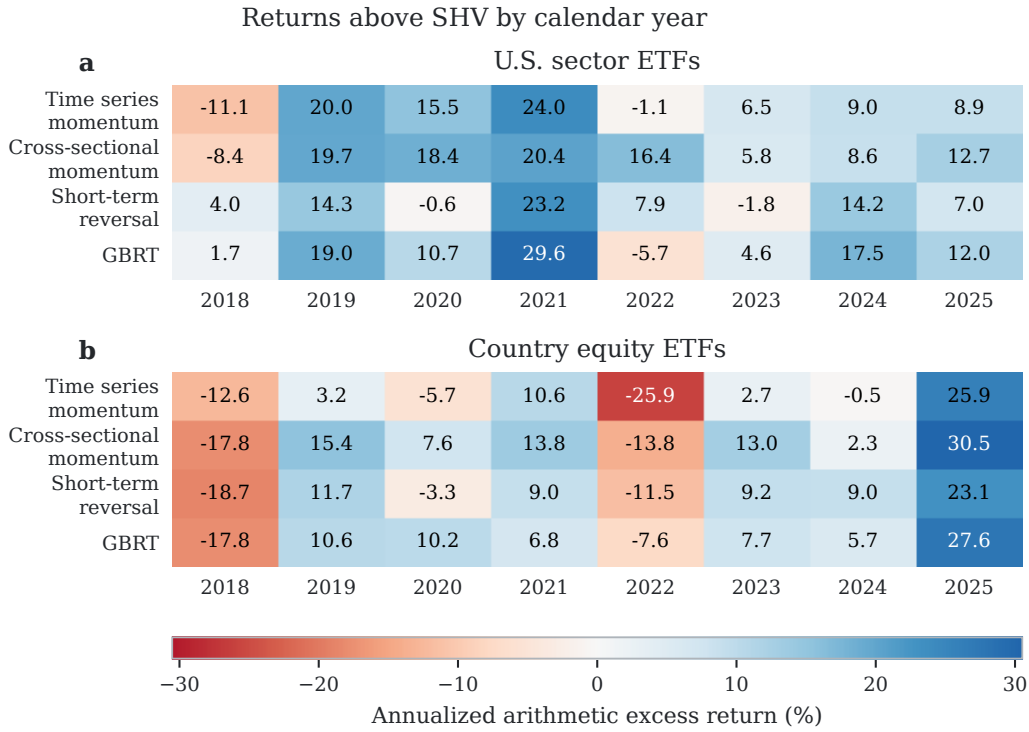


Figure 12: Baseline performance by calendar year as defined in Equation 9, expressed in percent per year. Each cell uses the original continuous portfolio path after costs and subtracts the aligned daily SHV return. Cells for the regression model average five individual seed values. Both subplots use the same color scale centered at zero. The annual slices describe temporal variation within the historical evaluation sample.

I.2 Accounting effects by seed

Figure 13 separates the five fixed seeds in each accounting comparison. Each cell subtracts two Sharpes for the same seed and ETF set, following the five defined subtraction orders. Averaging a column recovers the regression model's point estimate in Figure 6. The common color scale compares magnitudes; numerical labels preserve small cash and engine differences.

The seed mean can conceal differences in direction. Against the comparison portfolio after trading costs, the sector regression model has a negative Sharpe difference for every seed except 71; every country seed has a positive difference. Country engine differences change sign across seeds despite a positive mean. By contrast, cash, cost, and retroactive timing effects each have a consistent sign across all ten paths for the regression model.

Each row follows monthly fits that reuse the same integer seed. The five rows share market observations and one training specification; their spread describes the portfolio paths generated by those fits. The intervals in Figure 6 measure a different source of variation: they resample dates while retaining all five paths.

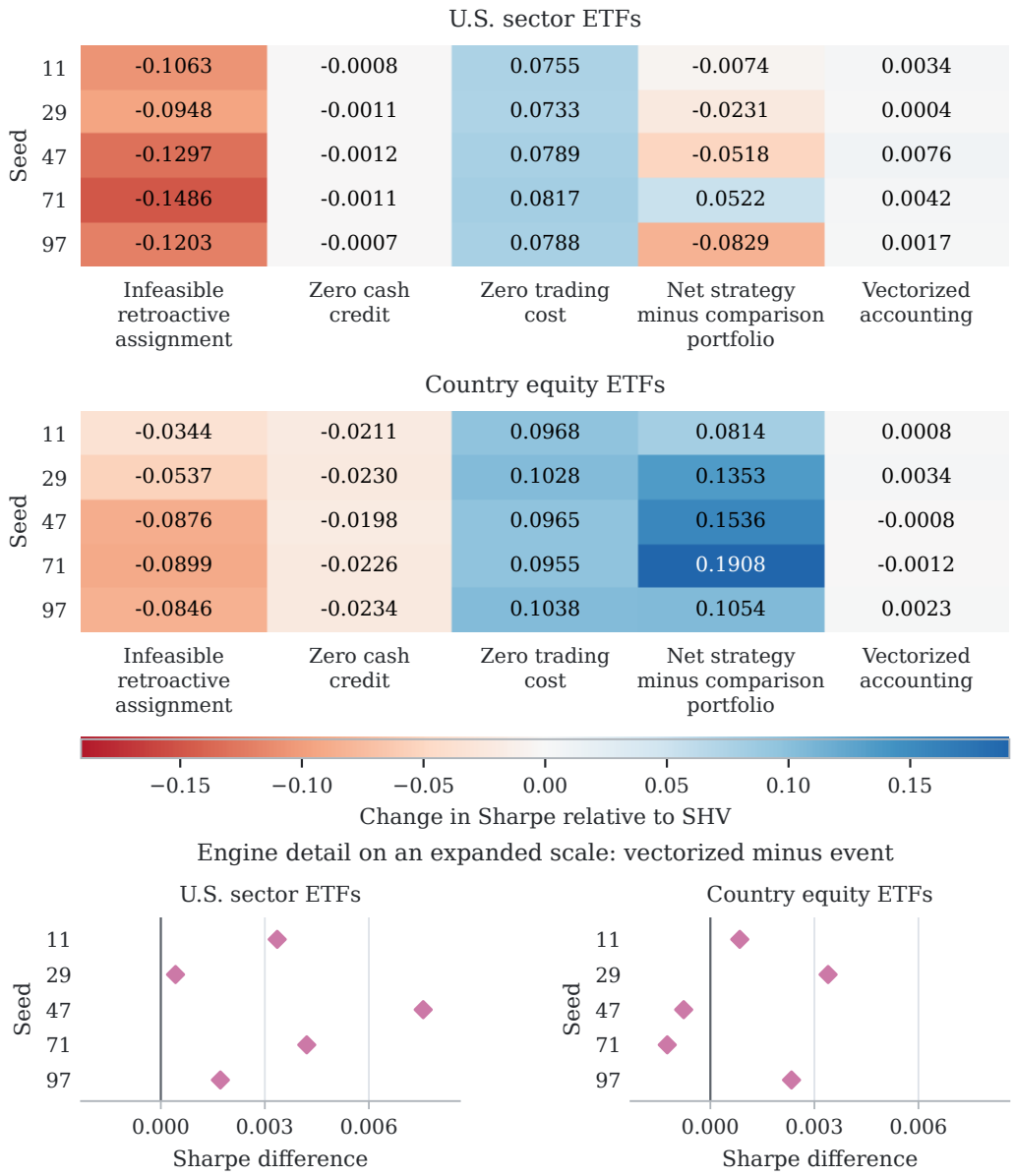


Figure 13: Changes in Sharpe relative to SHV within each of the five seeds for the regression model. Each matrix column subtracts baseline Sharpe from the displayed convention's Sharpe, except the comparison portfolio column, which reports strategy minus comparison portfolio, both after trading costs. The comparison portfolio matches that seed's target exposure. Retroactive assignment is infeasible and may also move the initial entry charge outside the evaluation window. Matrix colors share a scale centered at zero. The engine detail uses a separate horizontal scale, common to both ETF sets, to show the small positive and negative differences. All marks are point estimates.

I.3 Exposure and trading

Table 8 distinguishes risky exposure, trading activity, and the return deducted for costs along each baseline path. Gross traded notional sums purchases and sales relative to portfolio value immediately before each trade, counting both sides of a rotation and changes in total risky exposure. The cost column is $252 \, \mathrm{mean}(d_t)$, where $d_t = r_t^{\mathrm{gross}} - r_t^{\mathrm{net}}$ is the event engine's cost deduction. Both returns use the previous close's portfolio value as their denominator. This arithmetic deduction measures costs incurred along the observed path; a separate rerun with zero modeled cost also changes subsequent wealth and trade sizes. The distinction depends on when holdings, trades, and costs enter portfolio valuation [Boyd et al., 2017].

Table 8: Baseline mean risky exposure, annual gross traded notional as a multiple of portfolio value, and annualized arithmetic cost deduction. Entries for the regression model average five individual metrics. All paths use the same charge of 13 basis points on purchases and sales.

Fund set	Strategy	Mean risky exposure	Annual gross traded notional	Annualized cost deduction
United States sectors	Time series momentum	99.90%	4.39	0.57%
United States sectors	Cross-sectional momentum	99.90%	5.26	0.68%
United States sectors	Short-term reversal	77.73%	13.79	1.80%
United States sectors	Gradient boosting	98.74%	12.64	1.65%
Country equities	Time series momentum	88.61%	4.89	0.64%
Country equities	Cross-sectional momentum	99.90%	5.20	0.68%
Country equities	Short-term reversal	70.88%	13.91	1.81%
Country equities	Gradient boosting	89.26%	13.13	1.71%

The mean annual gross traded notional for gradient boosting is more than twice that of cross-sectional momentum in both ETF sets. At the same flat trading charge, gradient boosting incurs larger cost deductions. For country ETFs, it also has lower mean risky exposure. These return and Sharpe comparisons combine differences in exposure, allocation, and trading over the full historical window. Estimating capacity or market impact would require trading size, liquidity, and execution data.

J Evidence summary

Evidence	Observed property	Further evidence needed
Exact daily decomposition	Net excess over cash equals four arithmetic components, daily and within each joint resample.	A causal design, a geometric linking rule, or implementable portfolios for those respective interpretations.
Checks on allocation records	Records have valid decision times, complete symbol coverage, and admissible target weights.	Generator information checks, execution alignment, cash observations, and tests of future target generation.
Comparison under common targets	The accounting convention changes while the allocation records stay fixed.	Broader cases for a general direction of bias, generator checks for information timing, and new targets for a different strategy.
Conditional comparison portfolio	The basket after trading costs shares the strategy's target exposure path.	A separately defined benchmark for an independent comparison, or a causal design for attribution of skill.
Comparison of two engines	Two stated accounting conventions produce a measurable return difference.	External accounting oracles for correctness and a wider implementation sample for typical error.
Simulated state failures	Simulated order tests reject stale writes, uncertain retries, and unresolved replacements for the same symbol.	Account exposure checks, delivery tests with concurrent clients, authenticated transport, venue trials, and deployment controls.

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